Identifying Noncooperative Collusion with Imperfect Monitoring

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Motivation

- Traditional view point on oligopoly market is that firms mainly engage in Cournot output competition or Bernard price competition. Government do not need to interfere the market;

- Recently there are bunches of literatures on frims' noncooperative collusion behavior. They are mainly focus on the question: what is the maximal profit which can be supported by a noncooperative collusion equilibrium in different context.

- My question is this paper: **May government policy makers identify firms' noncooperative collusion behavior by only observing the market price?**

Model

$$log(Q_t) = \alpha_0 + \alpha_1 log(p_t) + \epsilon_t \tag{1}$$

$$Q_t = \sum_{i=1}^{n} x_{it}, \quad \epsilon_t \sim N(0, \sigma^2)$$

Where x_{it} is firm $i's$ production in period t, marginal cost c for each firm is common knowledge in our model. We consider the trigger price strategy described in Green and Porter (1984). Two important parameters: trigger price $\bar{p}$ and the length of punishing periods $M-1$.

$$x_{it} = \begin{cases} y & \text{if t is normal} \\ z & \text{if t is reversionary} \end{cases}$$

Where $z = \frac{c^{\alpha_1}}{ne^{\frac{\sigma^2}{2\alpha_1} - \alpha_0}(1+\frac{1}{n\alpha_1})^{\alpha_1}}$ is the Cournot output level under the expected demand function.

Determing Equilibrium Normal Output Level

We define:

$$\gamma_i(r) = E_\epsilon \pi_i[r, p[r + (n-1)y, \epsilon]]$$

$$\delta_i = E_\epsilon \pi_i[z, p(nz, \epsilon)]$$

$$V_i(r) = \gamma_i(r) + \beta Pr[\bar{p} \leq p(r + (n-1)y, \epsilon)] V_i(r)$$

$$+ Pr[p(r + (n-1)y, \epsilon) < \bar{p}][\sum_{t=1}^{M-1} \beta^t \delta_i + \beta^M V_i(r)]$$

$$V_i(r) = \frac{\gamma_i(r) + \Phi[\frac{log(r+(n-1)y) - \alpha_0 - \alpha_1 log(\bar{p})}{\sigma}] \frac{\beta - \beta^M}{1-\beta} \delta_i}{1 - \beta + (\beta - \beta^M)\Phi[\frac{log(r+(n-1)y) - \alpha_0 - \alpha_1 log(\bar{p})}{\sigma}]}$$

The first-order condition is:

$$\frac{dV_i(r)}{dr}|_{r=y} = 0$$

We denote the solution as:

$$y = y(\alpha_0, \alpha_1, \bar{p}, M, \sigma)$$

Identification Strategy

Our goal: By only observing market price p_t, we try to consistantly estimate $\alpha_0, \alpha_1, \bar{p}, M, \sigma$.

In our model, there are only two aggregate output levels:

$$Q_1 = ny \text{ or } Q_2 = nz$$

Since:

$$log(p_t) = \frac{log(Q_t) - \alpha_0}{\alpha_1} - \frac{\epsilon_t}{\alpha_1}$$

Then for each period t, $log(p_t)$ is generated from either of the following two process:

$$log(p_t) = \mu_1 + u_t \text{ or } log(p_t) = \mu_2 + u_t$$

Where $\mu_1 = \frac{log(ny)-\alpha_0}{\alpha_1}$, $\mu_2 = \frac{log(nz)-\alpha_0}{\alpha_1}$ and u_t are $i.i.d.$ from $N(0, \frac{\sigma^2}{\alpha_1^2})$.

Identification Strategy

Markov Switching Regression:

$$log(p_t) = \beta_{s_t} + u_t \tag{2}$$

$$\beta_0 = \mu_1$$

$$\beta_1 = \beta_2 = \cdots = \beta_{M-1} = \mu_2$$

Where $u_t \sim i.i.d.N(0, \sigma'^2)$ and $\sigma'^2 = \frac{\sigma^2}{\alpha_1^2}$. s_t is the outcome of an unobserved M-state Markov chain with state space=$\{0, 1, 2, \ldots, M-1\}$ wih transition matrix:

$$\mathbf{P} = \begin{bmatrix} 1-\Phi(\frac{\alpha_1(\bar{p}-\mu_1)}{\sigma}) & \Phi(\frac{\alpha_1(\bar{p}-\mu_1)}{\sigma}) & 0 & 0 & \ldots & 0 \\ 0 & 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 1 & 0 \\ 0 & 0 & 0 & \ldots & 0 & 1 \\ 1 & 0 & 0 & \ldots & 0 & 0 \end{bmatrix}$$

Identification Strategy

Now we briefly describe how we estimate the model for a given state size M by using the method from Hamilton (1989) with similar notations. Denote $p = \Phi(\frac{\alpha_1(\bar{p}-\mu_1)}{\sigma})$ and $\theta = (\mu_1, \mu_2, p, \sigma'^2)$.

$$f(log(p_t)|s_t = j; \theta) = \frac{1}{\sqrt{2\pi}\sigma'} exp\{-\frac{(log(p_t) - \beta_j)^2}{2\sigma'^2}\}$$

Denote:

$$\eta_t = \begin{bmatrix} f(log(p_t)|s_t = 0; \theta) \\ f(log(p_t)|s_t = 1; \theta) \\ \vdots \\ f(log(p_t)|s_t = M-1; \theta) \end{bmatrix}$$

$$\hat{\xi}_{t|t} = \frac{(\hat{\xi}_{t|t-1} \odot \eta_t)}{\mathbf{1'}(\hat{\xi}_{t|t-1} \odot \eta_t)} \tag{3}$$

$$\hat{\xi}_{t+1|t} = \mathbf{P} \cdot \hat{\xi}_{t|t} \tag{4}$$

Identification Strategy

The log likelihood function $\ell(\theta)$ from this algotithm is:

$$\ell(\theta) = \sum_{t=1}^{T} logf(log(p_t)|\theta) = \sum_{t=1}^{T} log[\mathbf{1}'(\hat{\xi}_{t|t-1} \odot \eta_t)]$$

$$\hat{\theta} = \arg\max \ell(\theta)$$

Choosing M:

Approach one: take the $M-1$-state model as the null and conduct a variety of tests of the validity of that specification as one way of seeing whether a M-state model is needed. Hamilton (1996) proposed a number of such tests.

Approach two: If we add one more state say from $M-1$ to M states, the total unknown parameters do not change in our model, so we might estimate M by:

$$\hat{M} = \arg \max_{1 \leq M \leq T} \ell(\hat{\theta}(M))$$

Identification Strategy

Now we have estimates $\hat{\theta} = (\hat{\mu}_1, \hat{\mu}_2, \hat{p}, \hat{\sigma'}^2)$ and $\hat{M}$. Recall that our goal is to estimate $\alpha_0, \alpha_1, \bar{p}, M,$ and σ. We can recover these parameters by solving the following system of equations:

$$\begin{cases} \frac{log(ny)-\alpha_0}{\alpha_1} = \hat{\mu}_1 \\ \frac{log(nz)-\alpha_0}{\alpha_1} = \hat{\mu}_2 \\ \Phi(\frac{\alpha_1(\bar{p}-\mu_1)}{\sigma}) = \hat{p} \\ \frac{\sigma^2}{\alpha_1^2} = \hat{\sigma'}^2 \\ M = \hat{M} \end{cases} \tag{5}$$

Where y and z depend on parameters $\alpha_0, \alpha_1, \bar{p}, M,$ and σ with known functional forms. So we can eventually identify all the parameters in our original model.